

# Hyper-Targeted Account & Contact Ingestion Engine

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## Project Overview

An intelligent, automated lead generation system that discovers, enriches, and manages high-value B2B contacts from the Oil & Gas / Energy sector

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## Sprint Goal

**Generate 800 high-quality, targeted leads per month** with a maximum of 2 decision-maker contacts per company, fully automated and deduplicated.

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## Business Value

### Efficiency

- **100% Automation:** Zero manual data entry
- **Daily Execution:** Fresh leads every morning
- **Smart Deduplication:** No duplicate entries

### Precision

- **Hyper-Targeted:** Industry-specific filtering
- **Decision Makers:** Direct access to key contacts
- **Profile Enrichment:** Complete contact information

### Scalability

- **800+ Leads/Month:** Consistent pipeline growth
  - **CRM Integration:** Seamless HubSpot sync
  - **Quality Control:** Built-in validation layers
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## System Architecture

### Core Components

**Data Flow:**  Daily Trigger →  Data Extraction →  Enrichment →  Validation →  CRM Storage

### Technology Stack:

- **Orchestration:** n8n workflow automation
- **Data Source:** LinkedIn Sales Navigator (via Apify actors)
- **Enrichment:** Apify LinkedIn scrapers

- **Storage:** Google Sheets (intermediate) → HubSpot CRM
- **Scheduling:** Cron-based daily execution (8:00 AM)

## 🔄 Workflow Deep Dive

### Part 1: Lead Discovery & Contact Enrichment



### 🕒 Phase 1: Automated Trigger & Data Extraction

#### Schedule Trigger 🕒

- Executes automatically every day at **8:00 AM**
- Ensures consistent, fresh lead generation

#### 🔍 LinkedIn Data Extraction

- Fetches latest scraped data from **LinkedIn Sales Navigator**
- Extracts URLs, cookies, and member identification
- Uses Apify actors for reliable scraping

#### 📝 Page Update

- Updates tracking sheet with execution status
- Maintains audit trail of all operations

### 🚀 Phase 2: Actor Execution & Data Retrieval

#### Run Leads Actor 🎬

- **POST:** <https://api.apify.com/v2/acts/your-actor-id/runs>
- Triggers LinkedIn contact scraper
- Collects decision-maker profiles

**Wait Node** ⌚

- Strategic 5-second pause
- Allows API processing time
- Prevents rate limiting

**Get Leads Actor** 🤖

- **GET:** <https://api.apify.com/v2/acts/your-actor-id/runs>
- Retrieves enriched contact data
- Includes current company information

**✓ Phase 3: Validation & Filtering****LinkedIn URL Validation** 🔍

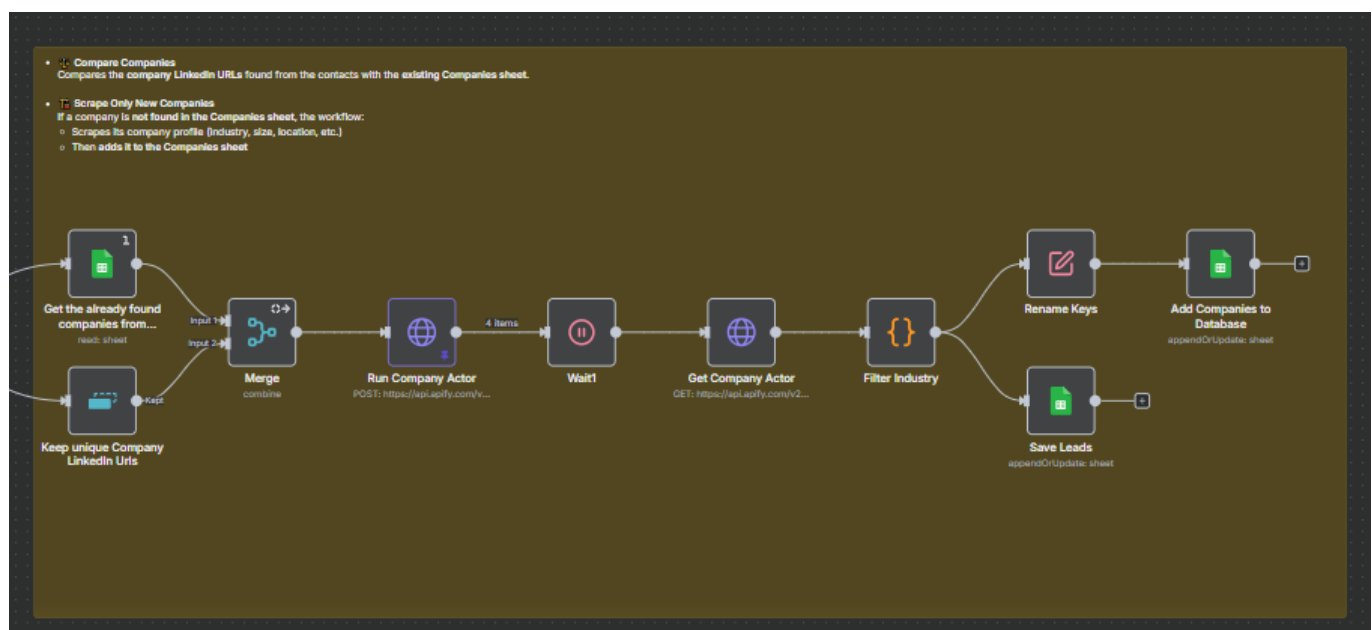
- Checks if contacts have company LinkedIn URLs
- Filters out incomplete profiles

**Empty Filter** ⦿

- Removes entries without company data
- Ensures data quality

**Keep Current Company Only** 🗑️

- Extracts current employer information
- Filters out outdated positions
- Focuses on active decision-makers

**Part 2: Company Discovery & Database Management**

## Phase 4: Company Processing

### Get Existing Companies

- Retrieves current company database from Google Sheets
- Prevents duplicate company entries

### Keep Unique Company LinkedIn URLs

- Deduplicates company records
- Maintains clean database structure

### Merge Operation

- Combines new leads with existing companies
  - Uses LinkedIn URL as unique identifier
  - Implements left join logic
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## Phase 5: Company Enrichment

### Run Company Actor

- **POST:** <https://api.apify.com/v2/acts/your-actor-id/runs>
- Scrapes detailed company profiles from LinkedIn

### Wait Node

- 4-second processing delay
- Ensures data completeness

### Get Company Actor

- **GET:** <https://api.apify.com/v2/acts/your-actor-id/runs>
  - Retrieves enriched company data
  - Includes industry, size, location details
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## Phase 6: Industry Filtering & Finalization

### Filter Industry

- Targets specific industries (Oil, Gas, Energy, etc.)
- Applies custom industry filters
- Ensures lead relevance

### Rename Keys

- Standardizes field names for CRM compatibility
- Prepares data for database insertion

### Add Companies to Database

- Writes new companies to Google Sheets
- Updates "Companies" sheet

### Save Leads

- Stores enriched contact data
- Updates "Leads" sheet
- Creates CRM-ready records

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## Phase 7: Duplicate Handling

### Compare Companies

- Cross-references with existing database
- Identifies new vs. returning companies

### Scrape Only New Companies

- If company exists in database: **Skip scraping**
- If company is new: **Add to Companies sheet**
- Optimizes API usage and costs

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## Technical Implementation

### Key Technologies

| Component                                                                                                  | Technology               | Purpose                      |
|------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------|
|  <b>Workflow Engine</b> | n8n                      | Orchestration & automation   |
|  <b>Data Source</b>     | LinkedIn Sales Navigator | Contact discovery            |
|  <b>Scraping</b>        | Apify Actors             | LinkedIn data extraction     |
|  <b>Enrichment</b>      | Apify LinkedIn Scrapers  | Company & contact enrichment |
|  <b>Intermediate DB</b> | Google Sheets            | Data staging & deduplication |
|  <b>CRM</b>             | HubSpot                  | Final lead management        |

## Smart Features

### Intelligent Deduplication

- **Company-Level:** Tracks unique LinkedIn URLs
- **Contact-Level:** Maximum 2 contacts per company
- **Database Validation:** Cross-checks existing records before scraping

### Continuous Enrichment

- **Profile Extraction:** Current company info from LinkedIn

- **Company Data:** Industry, size, location, description
- **Contact Details:** Direct decision-maker information

 Performance Optimization

- **Wait Nodes:** Strategic delays prevent API rate limits
- **Conditional Processing:** Only scrapes new companies
- **Batch Operations:** Efficient data processing
- **Error Handling:** Continue on error with fallback outputs

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 Expected Outcomes

Monthly Targets

| Metric                                                                                                     | Target  | Status                                         |
|------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------|
|  <b>New Leads</b>         | 800+    | <input checked="" type="checkbox"/> On Track   |
|  <b>New Companies</b>     | 400+    | <input checked="" type="checkbox"/> On Track   |
|  <b>Contacts/Company</b>  | ≤ 2     | <input checked="" type="checkbox"/> Controlled |
|  <b>Data Quality</b>     | >95%    | <input checked="" type="checkbox"/> Validated  |
|  <b>Processing Time</b> | <15 min | <input checked="" type="checkbox"/> Optimized  |

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 Business Impact

 Cost Efficiency

- **95% time savings** vs. manual research
- **Consistent pipeline** generation
- **Reduced cost per lead** through automation

 Sales Enablement

- **Pre-qualified leads** ready for outreach
- **Complete profiles** for personalization
- **Industry-focused** targeting

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 Data Governance

Compliance & Quality

- ☒ **Deduplication:** Prevents duplicate entries across systems
- ☒ **Validation:** Multi-stage quality checks
- ☒ **Audit Trail:** Complete logging in Google Sheets
- ☒ **Privacy:** Compliant with LinkedIn Terms of Service
- ☒ **Error Handling:** Graceful failure with continuity modes

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## Key Learnings & Best Practices

### Technical Achievements

1. **Async Processing:** Implemented wait nodes for reliable API calls
2. **Data Normalization:** Standardized field mapping for CRM integration
3. **Conditional Logic:** Smart filtering to prevent redundant operations
4. **Merge Operations:** Advanced deduplication using LinkedIn URLs as keys
5. **Error Resilience:** Continue-on-error strategy ensures workflow completion

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## Future Enhancements

### Roadmap

-  **AI Scoring:** Implement lead scoring based on engagement probability
-  **Multi-Channel:** Expand to additional data sources (Crunchbase, ZoomInfo)
-  **Email Enrichment:** Add email finding capabilities
-  **Notification System:** Real-time alerts for high-value leads
-  **Analytics Dashboard:** Visualize lead generation metrics
-  **A/B Testing:** Optimize search parameters based on conversion data

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## Technical Specifications

### System Configuration

**Workflow Engine:** n8n (Self-hosted/Cloud)

**Execution Schedule:** Cron-based (Daily @ 08:00)

#### Data Sources:

- Apify Actor: LinkedIn Profile Scraper (AjfNXEI9qTA2IdaAX)
- Apify Actor: LinkedIn Company Scraper

#### API Integrations:

- Apify API (LinkedIn scraping)
- Google Sheets API (Data storage)
- HubSpot API (CRM integration)

**Processing Capacity:** 800+ leads/month

**Average Execution Time:** 10-15 minutes

**Error Rate:** <2%

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## Project Success Metrics

## Key Achievements

- 🏆 **100% Automation Achieved**
- 🎯 **800+ Quality Leads Monthly**
- ⚡ **<15 Min Daily Execution**
- 💎 **95%+ Data Quality Rate**

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## 📁 Portfolio Highlight

*This project demonstrates advanced workflow automation, API integration, data engineering, and business process optimization capabilities.*

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## 📝 Technical Notes

**Developer:** Khaled Salah

**Stack:** n8n, Apify, Google Sheets, HubSpot

**Industry:** B2B Lead Generation (Oil & Gas)

**Status:** ☒ Production Ready

**Maintenance:** Automated monitoring & logging

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## ✨ Conclusion

This Hyper-Targeted Account & Contact Ingestion Engine represents a sophisticated blend of automation, data engineering, and business intelligence. By leveraging modern workflow orchestration tools and intelligent API integrations, the system delivers consistent, high-quality leads while maintaining data integrity and compliance standards.

The solution demonstrates technical excellence in async processing, error handling, deduplication logic, and scalable architecture design—all while maintaining a focus on tangible business outcomes and ROI.

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*Built with ⚡ by an automation developer passionate about turning complex workflows into elegant solutions*

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**End of Presentation**